

Amplificator Operational Rail to Rail

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Cuprins

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1) Specifications:

- a. Rail to Rail Input
- b. Rail to Rail output (within 90mv)
- c. Wide Bandwidth: 50MHz GBW
- d. High Slew Rate: 25V/us (10k load)
- e. Unity Gain Stable
- f. High output current: 10 mA
- g. Open Loop Voltage Gain: >80 dB
- h. Closed Loop Output Impedance @ G=0 dB: < 10 mOhm
- i. Supply Voltage 1.8 V
- j. Low open loop output resistance

2) Input stage

Input rail-to-rail operation requires that the N-channel and P-channel input pair to be placed in parallel, as shown in figure 1.

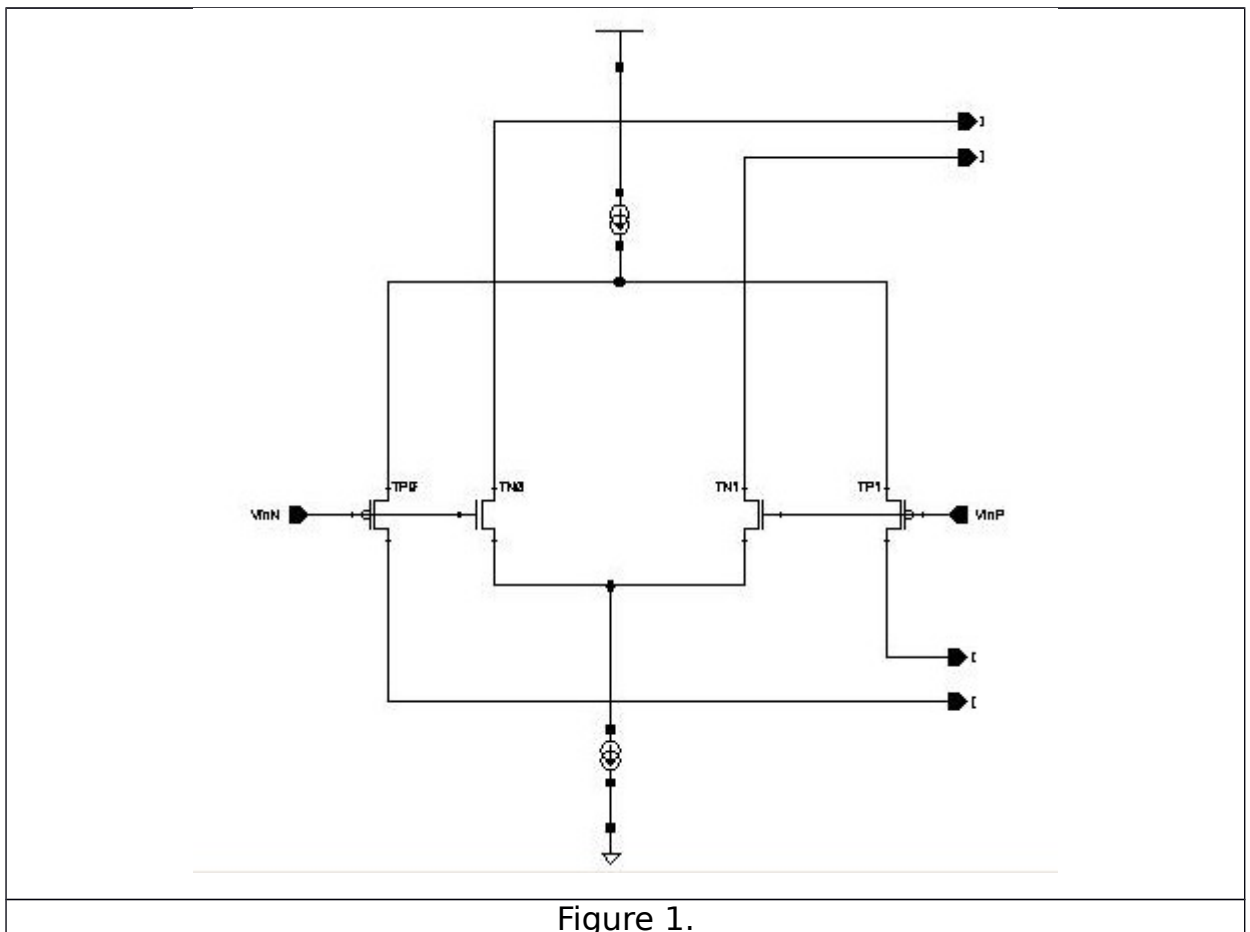


Figure 1.

Because there is a range of common input voltage, where the N-channel and P-channel stages work in parallel, the total G_m varies by a factor of two. To resolve this issue, many design techniques were developed. The one used in this project was proposed in [1], and is shown in figure 2. The purpose of the added circuit is to drain current

form the two current sources when the two input stages work in parallel, so that the overall G_m stays constant.

The G_m vs V_{CM} for the circuit in figure 1 and 2 is shown in figure 3. As we can see, with the added circuit, the transconductance variation is lower (approximately 10%). The dimensions of the transistors in figure 2 are:

- Nmos: $W = 12 \mu m$; $L = 360 nm$
- Pmos: $W = 75 \mu m$; $L = 360 nm$

The current sources are implemented using transistors and give a total current of $37.5 \mu A$ (for input pairs) + $6.25 \mu A$ (for the added circuit), see [1].

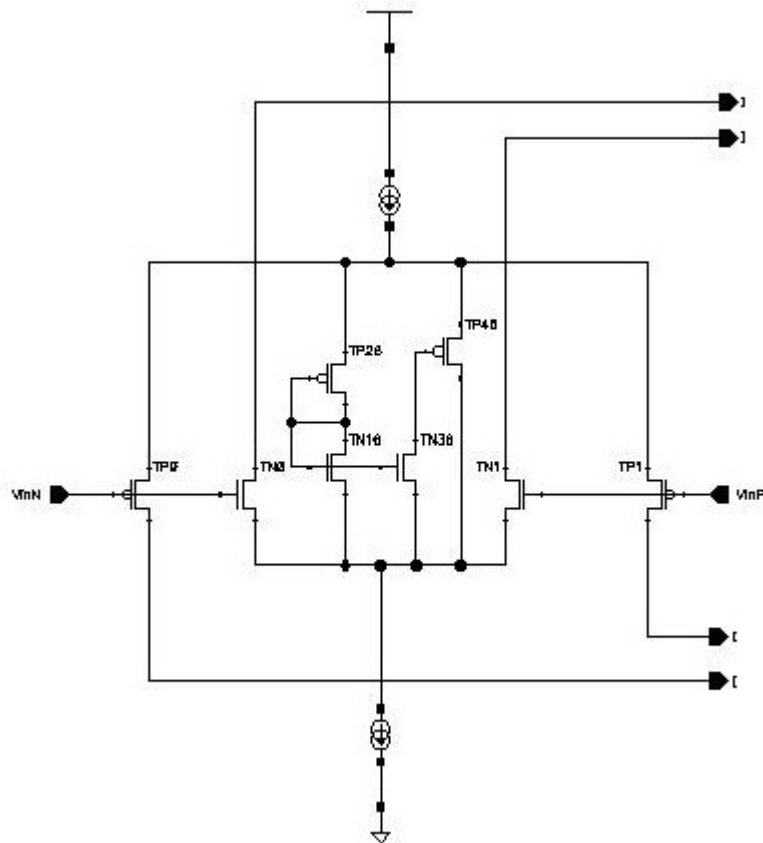
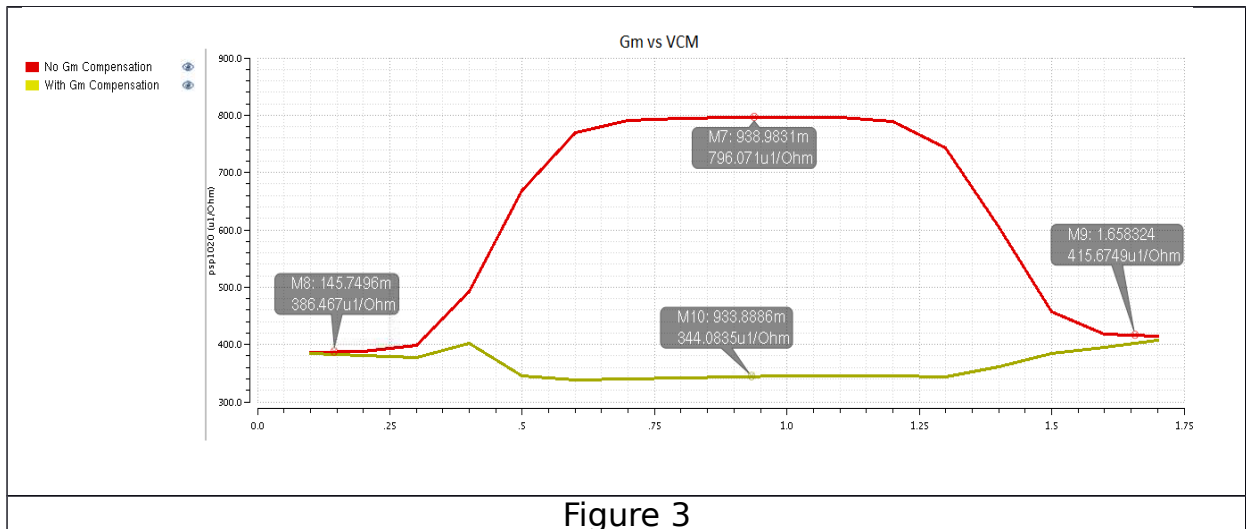


Figure 2



3) Summing circuit

The summing circuit is shown in figure 4. The current coming from the Nmos pair, goes into MP8 and MP9 sources, and the current from Pmos pair goes into MN8 and MN9 sources. To explain how this circuit works we will take three cases: Nmos pair on and Pmos pair off, Pmos pair on and Nmos pair off, Pmos pair and Nmos pair on.

a) Nmos pair on, Pmos pair off:

The current arriving into MP8 source, is forced to go through MP6, an through MP7, arriving at the drain of MP9. We will call this current I_{P1} . The current arriving at the source of MP9 is of the same polarity as I_{P1} . This means that the voltage at the drain of MP9 will increase. This voltage will be then amplified by MP14, MN14 transistors.

b) Nmos pair off, Pmos pair on:

The current arriving into MN8 source is forced to go through MP6 and then through MP7, arriving at the drain of MP9. We will call this current I_{N1} . At the same time the current that arrives at the source of MN9, reaches the drain of MN9. This current and I_{N9} cause the voltage at the drain of MP9 and MN9 to have the same polarities, bootstrapping MP13, MN13. This voltage will then be amplified by MP14, MN14 transistors.

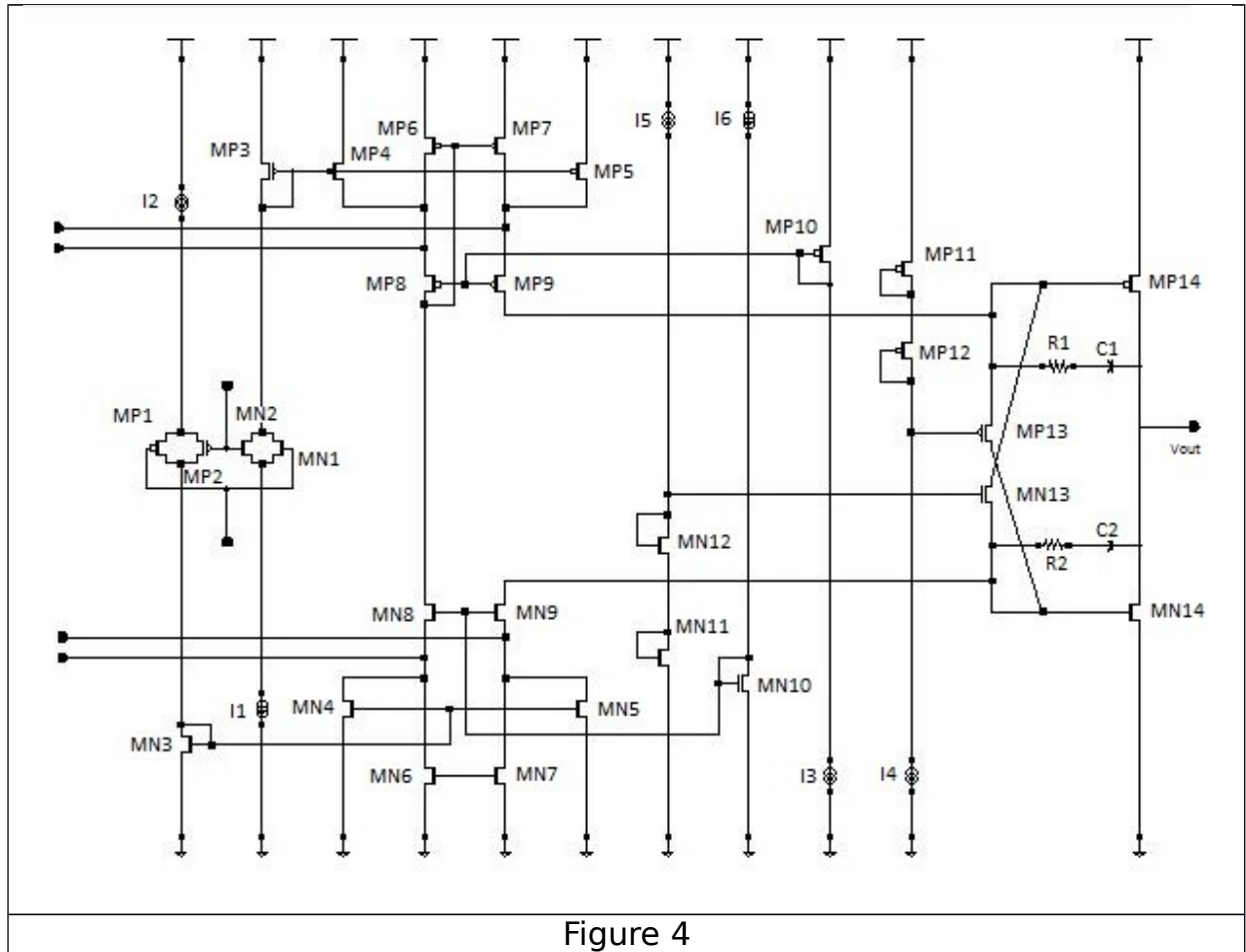


Figure 4

c) Nmos pair on, Pmos pair on:

Because the circuit is linear this situation can be analysed by considering the effect of the Nmos input stage, and then the effect of the Pmos input stage. This situations were analysed in a) and b). The only difference now is that the G_m of the input pairs is halved, to achieve a constant G_m over all common mode input voltage.

Transistors MP14 and MN14 form a class AB output stage. The quiescent current of these transistors is fixed with the help of translinear loops: MP11, MP12, MP13, MP14 and MN11, MN12, MN13, MN14. [2] .

MN6 and MN7 act as current sources (the bias voltage isn't shown in figure 4), and dictate the current through MN8, MN9, MP6, MP7, MP8, MP9, MP10, MN13, MP13. If the input common mode voltage changes from 0 to V_{DD} , the current through the transistors mentioned above changes affecting the frequency response of the amplifier. To alleviate this issue

transistors MN1, MN2, MP1, MP2, with gates connected to the input of the amplifier are used.

Suppose that V_{CM} is low, so only the Pmos input stage works. Then MN1, MN2 are off and MP1, MP2 are on, and I_2 flows through MN3 which forms a current mirror with MN4 and MN5. With careful sizing of MN5 and MN4, the current needs of the Pmos input pair can be satisfied by MN4 and MN5. The current through MN8 and MN9 is now the current supplied by MN6 and MN7.

In a similar manner we can show that when the Nmos pair is on, and the Pmos pair is off, the current needed by the former is supplied by MP4 and MP5, and the current through MN8 and MN9 is the current supplied by MN6 and MN7.

So by using MN1, MN2, MP1, MP2 transistors and their associated current mirrors, low variation of the quiescent current, of the transistors making the summing circuit, can be obtained, as shown in figure 5.

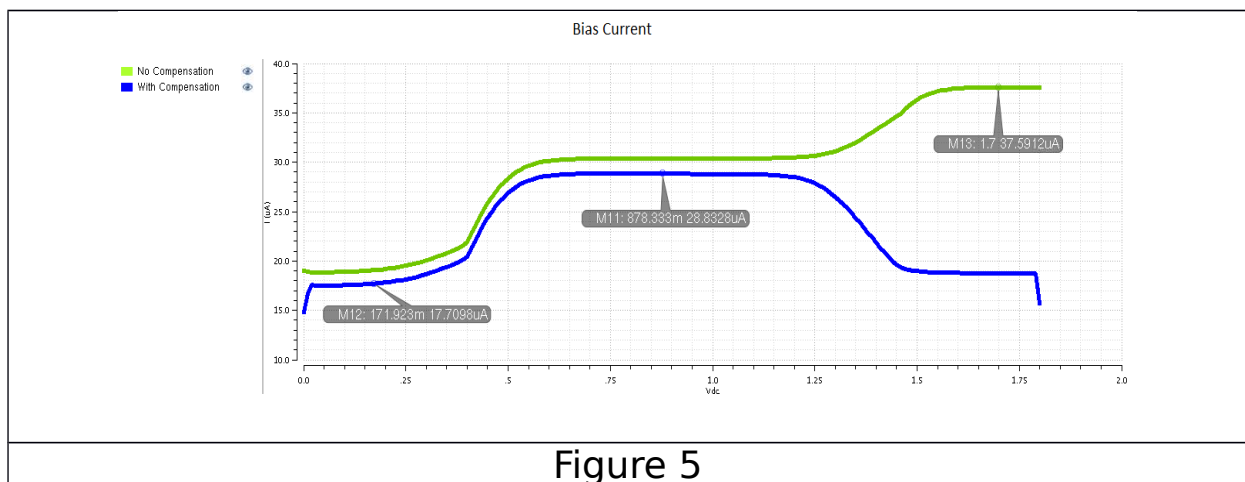


Figure 5

The size of the transistors that make the summing circuit are shown in table 1.

Transistor	Width(μm)	Length(μm)
MN1	12	0.36
MN2	12	0.36
MN3	11.25	1
MN4	11.25	1
MN5	11.25	1
MN6	11.25	1
MN7	11.25	1
MN8	22.5	0.18
MN9	22.5	0.18

MN10	2.2	3
MN11	5	0.36
MN12	5	0.36
MN13	10	0.36
MN14	24	0.36
MP1	75	0.36
MP2	75	0.36
MP3	54	1
MP4	54	1
MP5	54	1
MP6	54	1
MP7	45	1
MP8	108	0.18
MP9	108	0.18
MP10	10	1
MP11	26	0.36
MP12	26	0.36
MP13	52	0.36
MP14	90	0.36

The compensation network consists of a miller capacitor in series with a resistor for pole zero cancelation. The values of the resistors and capacitors are:

- $C = 400 \text{ fF}$
- $R = 5 \text{ k}\Omega$

4) Output stage

Because we need a low open loop impedance, a unity gain buffer is needed. To obtain high efficiency, a class AB output stage is required. The schematic is shown in figure 6 [3], [4].

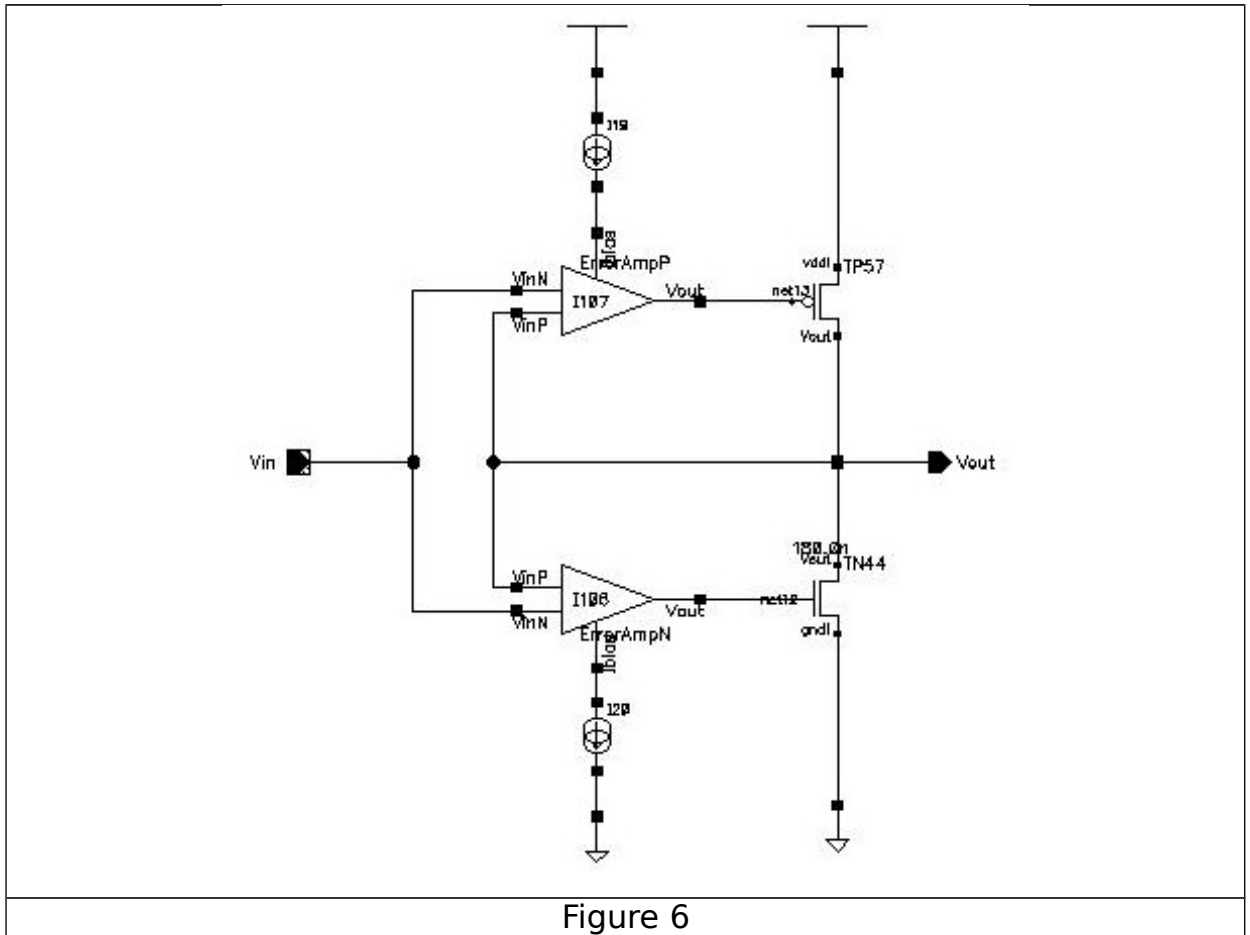


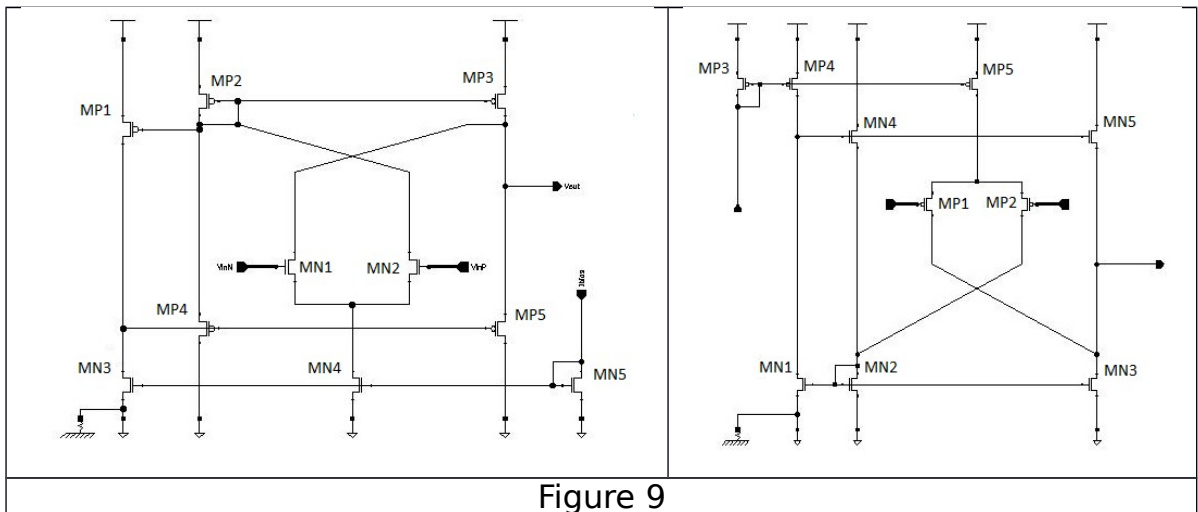
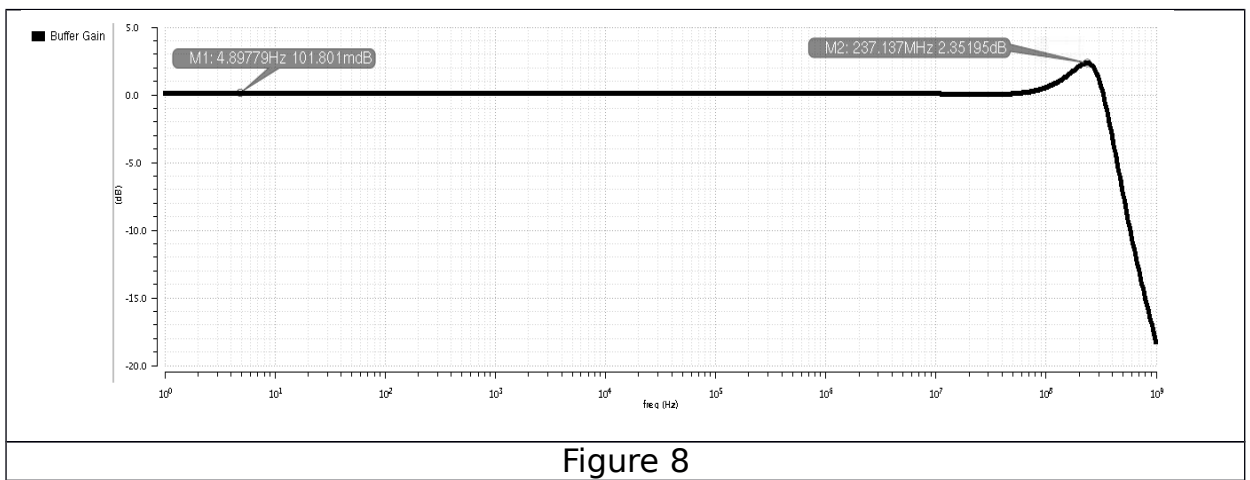
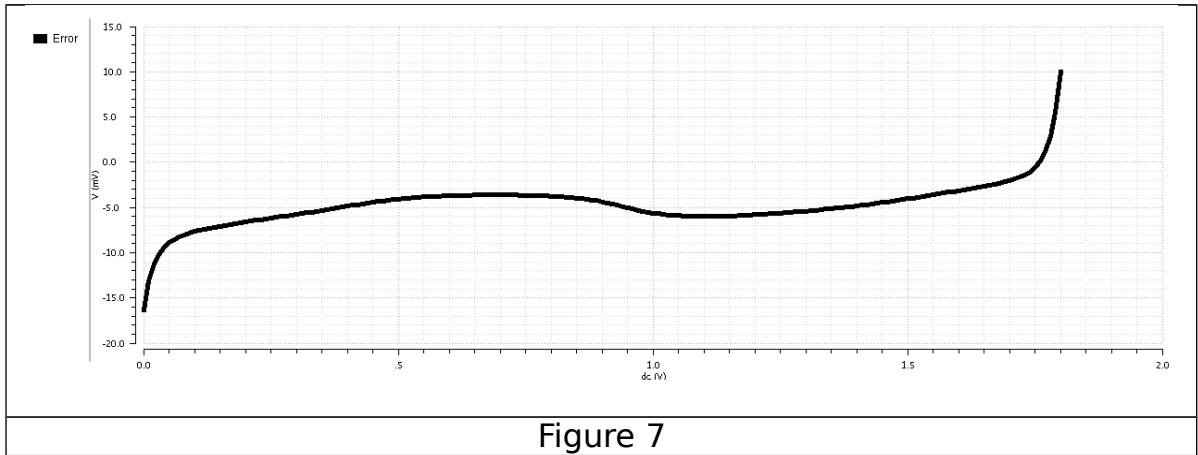
Figure 6

The low impedance and unity gain amplification is obtained with the help of negative feedback. If the error amplifiers have large open loop

gain, than $V_i = V_{out}$, and $R_{out} = \frac{r_{o57} \vee r_{o44}}{1+A}$, where

$A = A_{\text{error amp}} \cdot g_{m57,44} \cdot (r_{o44,57} \vee R_L)$, is the open loop gain, assuming perfect AB operation of the stage.

In figure 7 we show the difference between V_i and V_{out} . As we can see, it is far from 0. This happens because, the error amplifiers have low gain. The reason we choose a low gain is explained in [3]. The frequency response of the unity gain amplifiers is shown in figure 8. The schematic of the errors amplifiers is shown in figure 9.



The quiescent current through transistors TP57 and TN44 is fixed with the help of the error amplifiers. If we write KLV we obtain:

$V_{SG75} + V_{SG5} - V_{SG4} - V_{SG2} = 0$ and $V_{GS44} + V_{GS5} - V_{GS4} - V_{GS2} = 0$. From these equations we can obtain a relation between the size of the transistors,

bias current of the error amplifier and the quiescent current of the output transistors.

The size of the transistors in the output buffer are shown in table 2, for the ErrorAmpN, and in table 3 for the ErrorAmpP.

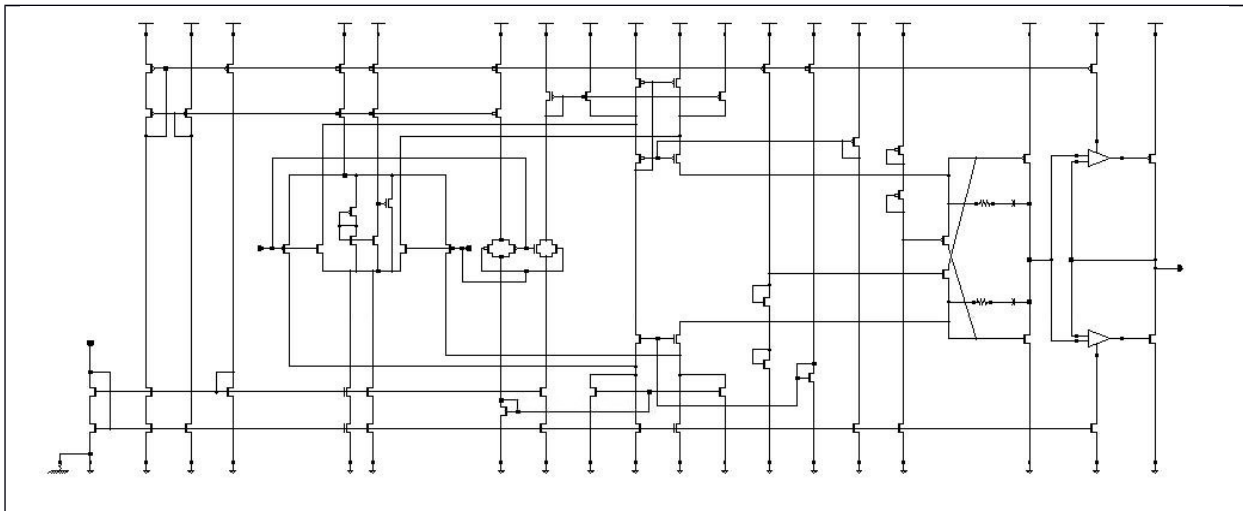
Table 2		
Transistor	Width(μm)	Length(μm)
MN1	60	0.36
MN2	60	0.36
MN3	3	2
MN4	36	2
MN5	18	2
MP1	54	0.36
MP2	432	0.36
MP3	432	0.36
MP4	1	0.36
MP5	1	0.36

Table 3		
Transistor	Width(μm)	Length(μm)
MN1	10	0.36
MN2	130	0.36
MN3	130	0.36
MN4	1	0.36
MN5	1	0.36
MP1	15	0.36
MP2	15	0.36
MP3	100	2
MP4	4.5	2
MP5	100	2

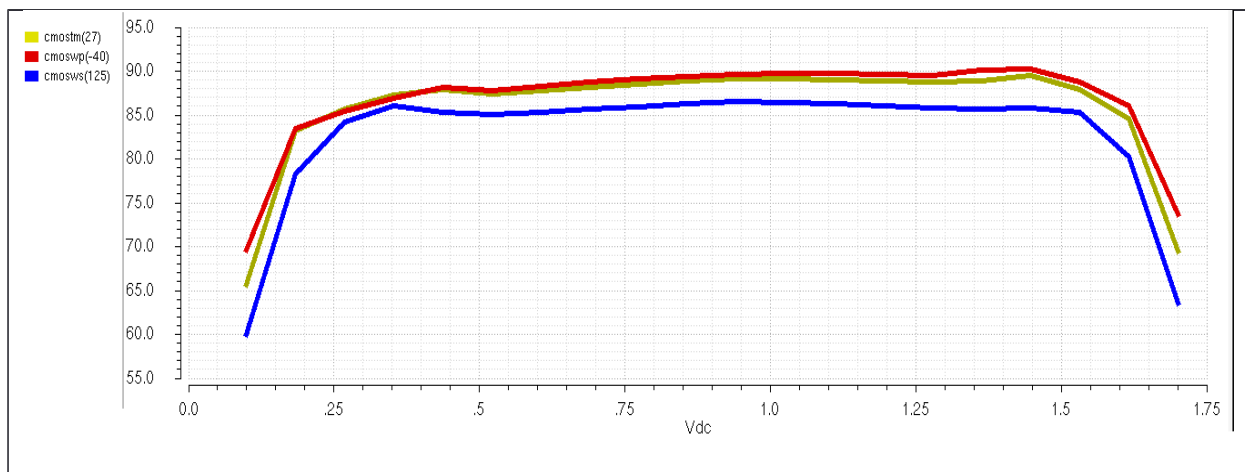
The dimensions of TP is : $W = 450 \mu m, L = 180 nm$, and of TN44 is:
 $W = 250 \mu m, L = 180 nm$

5) Overall design and simulations:

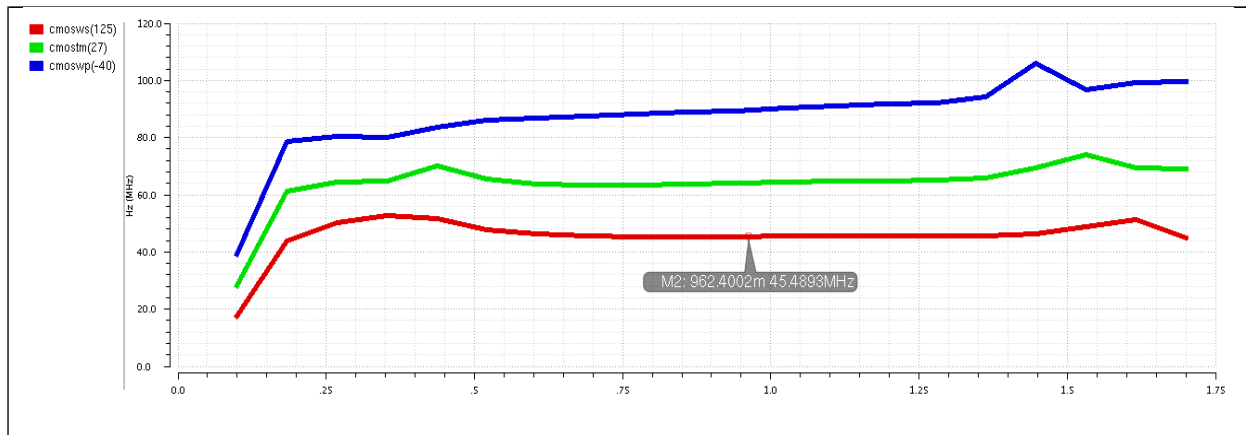
In figure 10 we show the full schematic of the operational amplifier, and in the next section some corner simulations:



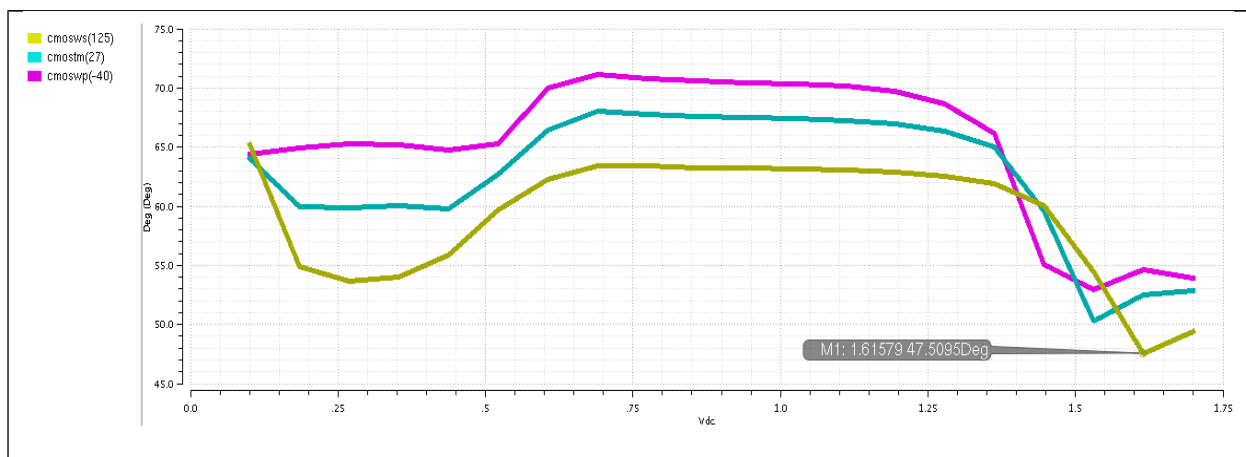
a) Gain vs input common mode voltage:



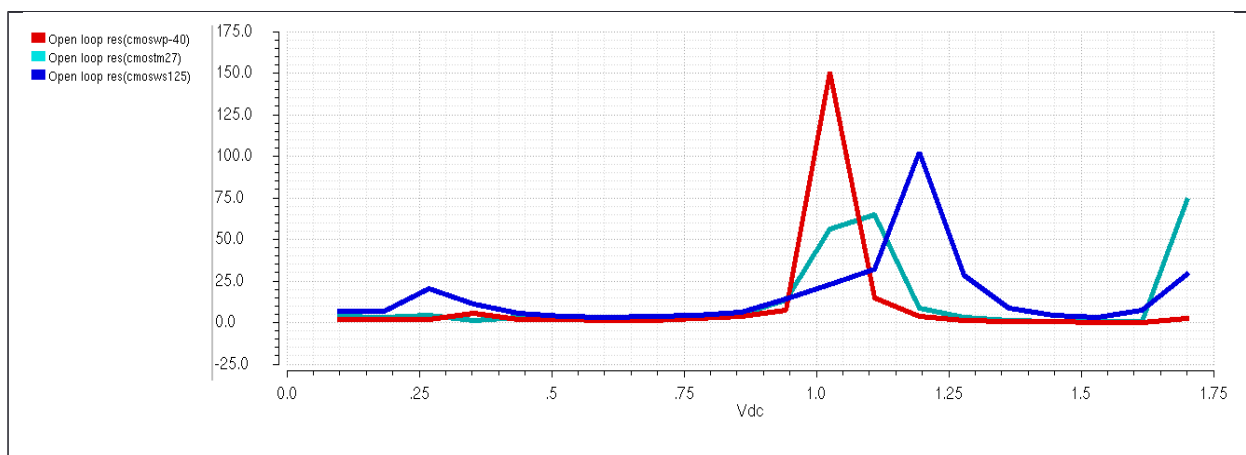
b) Unity gain frequency vs input common mode voltage:



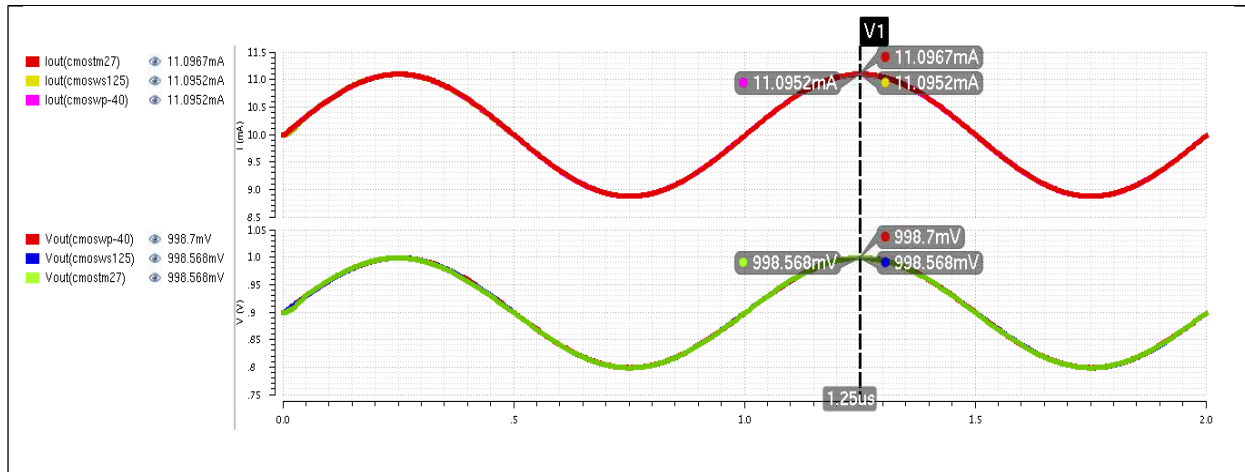
c) Phase Margin vs input common mode voltage:



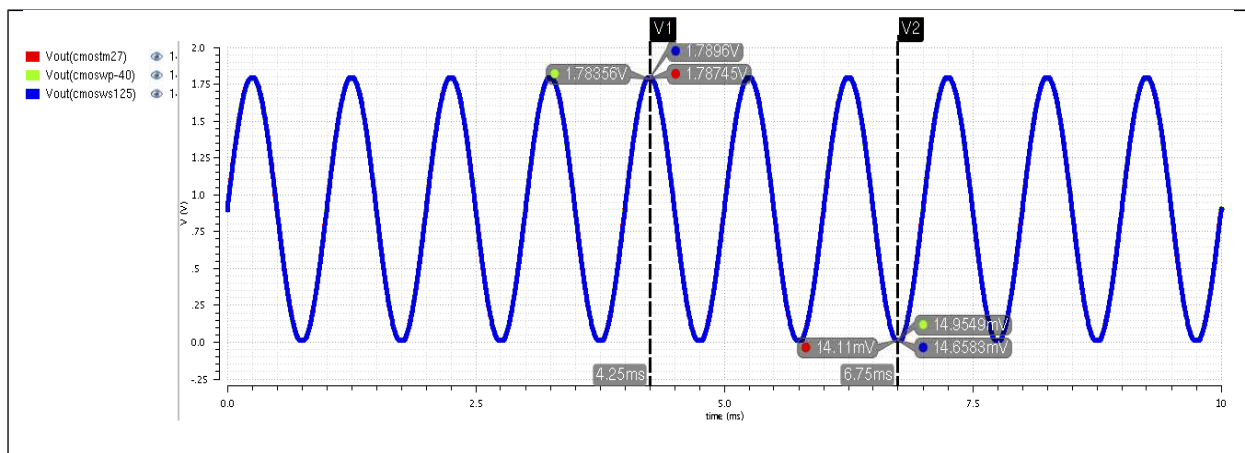
d) Open loop resistance vs input common mode voltage:



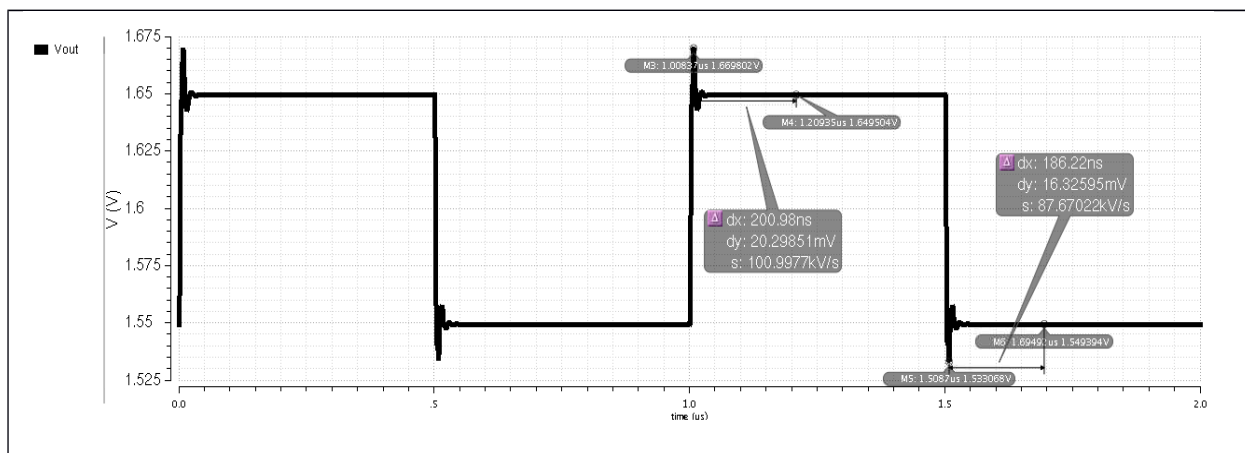
e) Output voltage and output current:



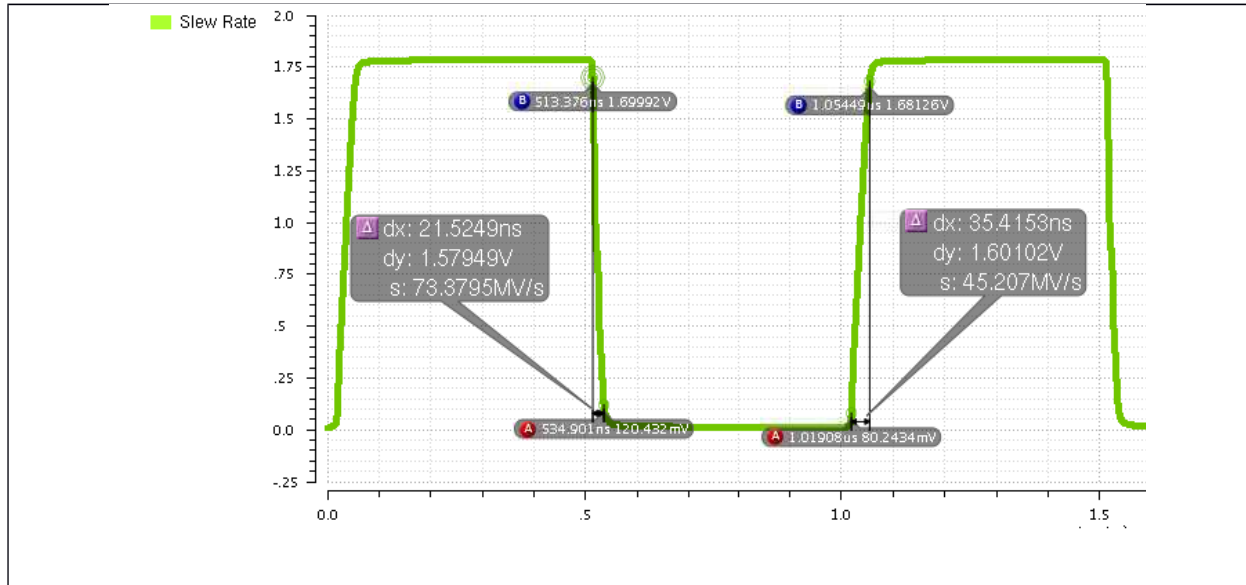
f) Full swing Vout:



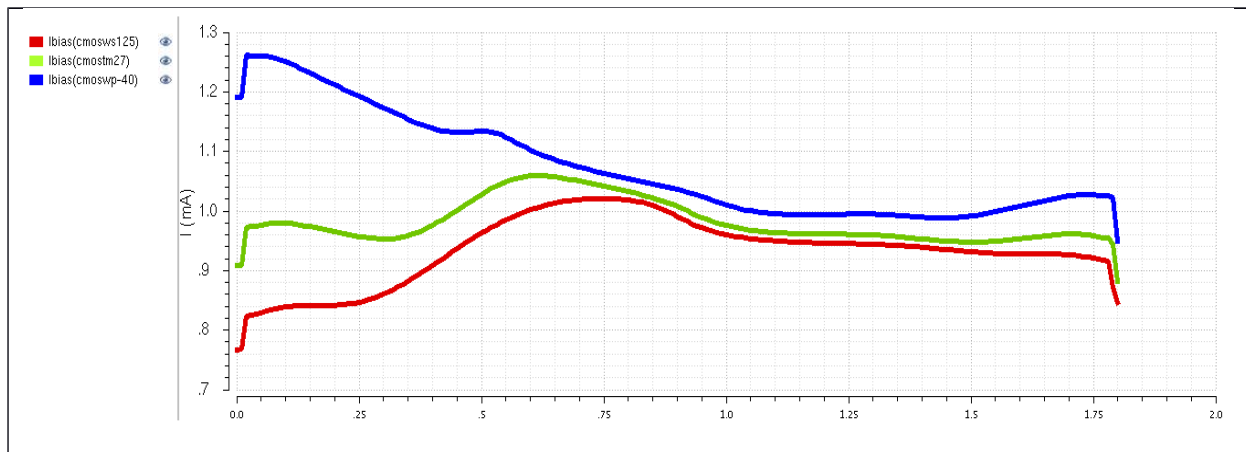
g) Small step response @ worst case PM:



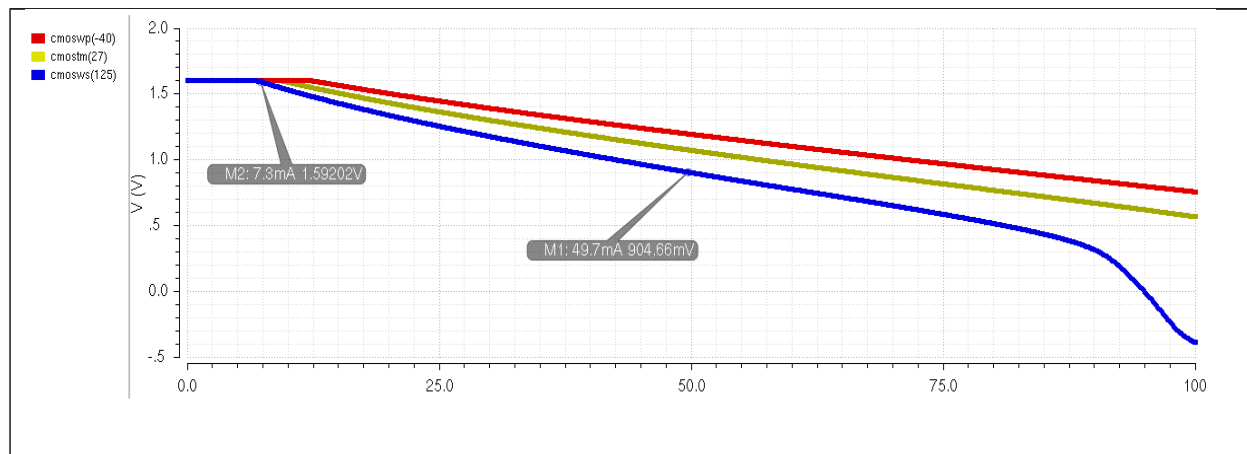
h) Slew Rate:



i) Bias current vs vs input common mode voltage:



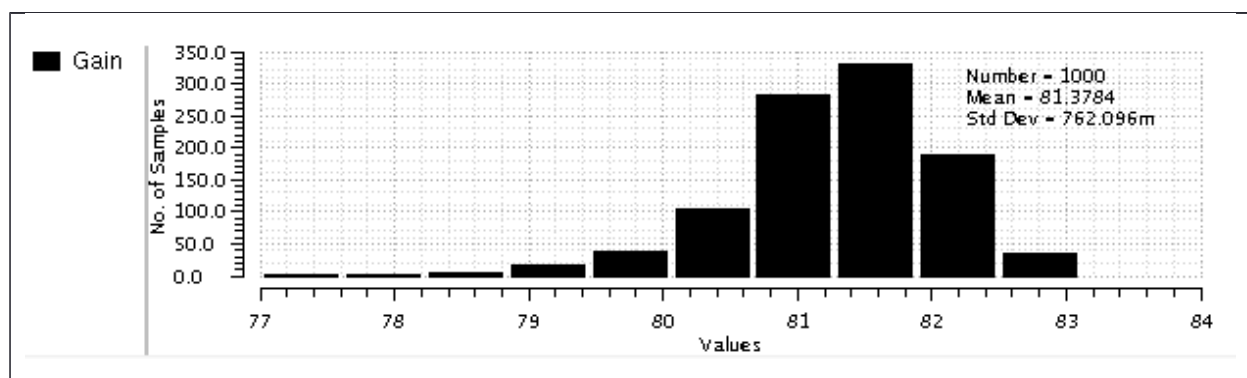
j) Maximum output voltage vs output current:



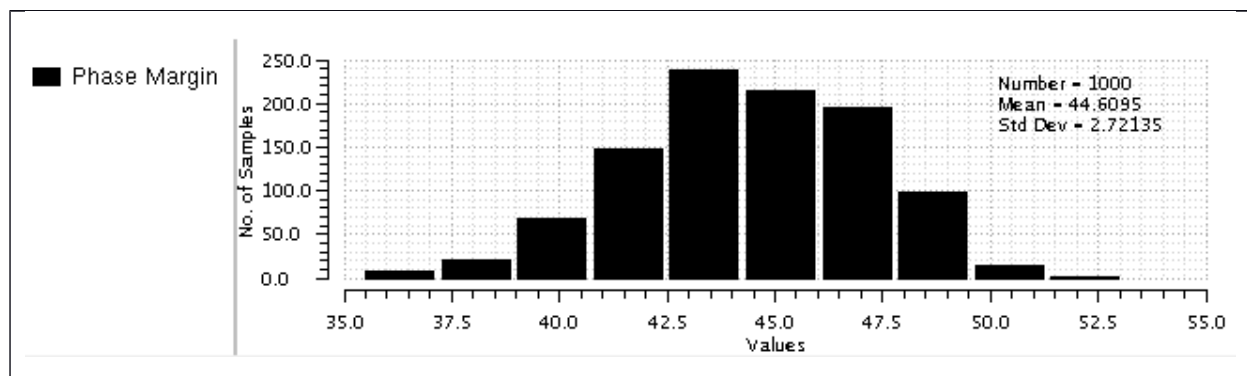
6) Monte Carlo analysis

Monte Carlo was done only for cmosws corner at 125°C, because how you can see, in this corner we obtained the worst performances.

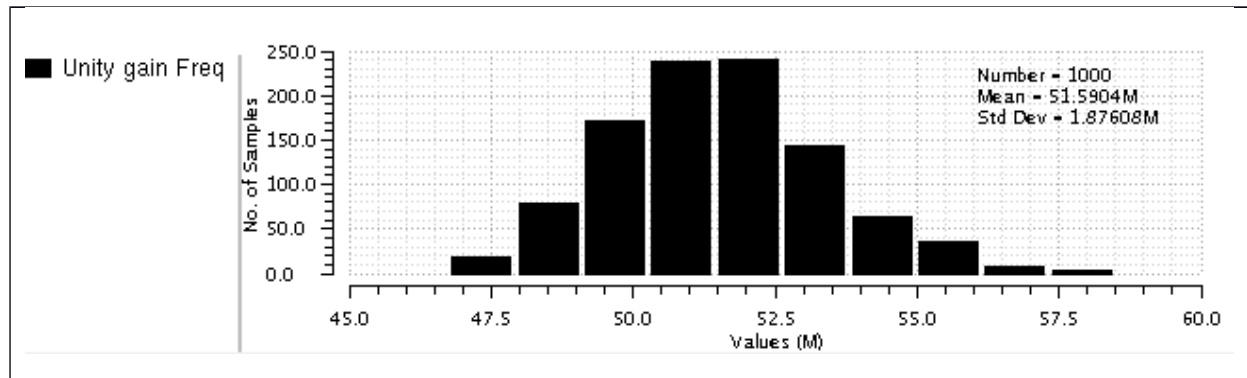
a) Gain (cmosws corner, 125°C, $V_{cm} = 1.6 \text{ V}$)



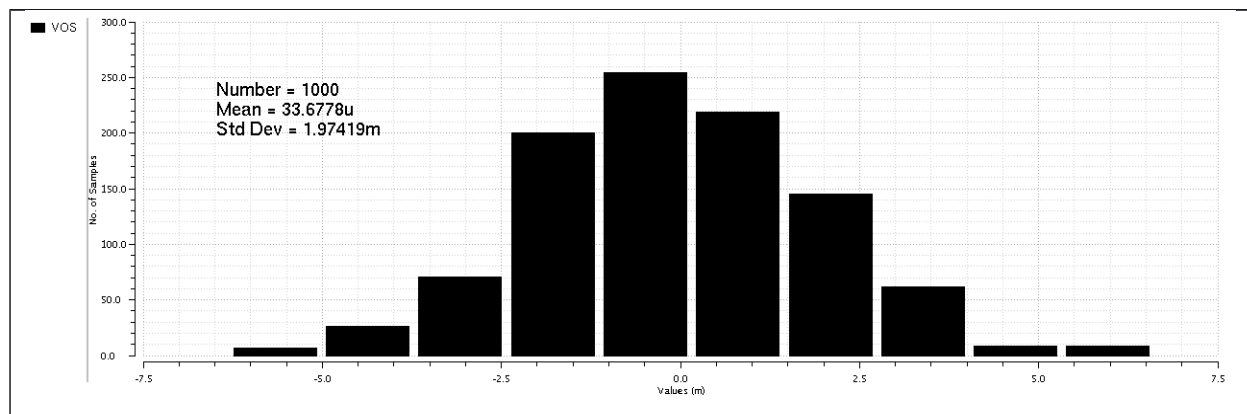
b) Phase Margin (cmosws corner, 125°C, $V_{cm} = 1.6 \text{ V}$)



c) Unity gain frequency (cmosws corner, 125°C, $V_{cm} = 1.6 \text{ V}$)



d) Offset Voltage (cmstm cornet, 27°C, $V_{cm}=900mV$)



Bibliography:

- 1) *Compact CMOS Constant g_m Rail-to-Rail Input Stage with g_m - Control by an Electronic Zener Diode* - Ron Hogervorst, John P. Tero, and Johan H. Huijsing, *IEEE JOURNAL OF SOLID-STATE CIRCUITS*, VOL33, NO 7 , JULY 1996
- 2) *Analog Design Essentials* - Willy M.C. Sansen, pag 346 - 348.
- 3) *Analysis and Design of Analog Integrated Circuits* 5th ed. - Paul R. Gray, pag 381 - 387
- 4) *Microelectronic Circuits* - Adel S. Sedra, Kenneth C. Smith 6th ed, pag 936 - 943